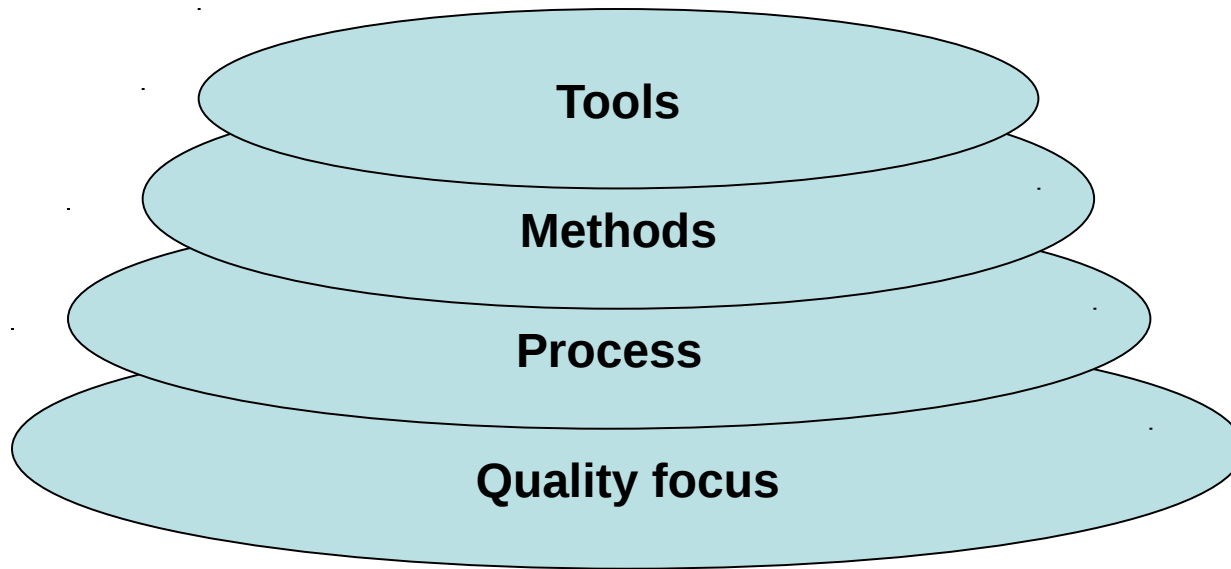

Software Development Lifecycle: Process Models

Layered approach to SE



Layered Approach to SE

- **Quality:** Focus on quality is the basis for developing software methodically.
- **Process layer:** Establishing a process for monitoring and control results in achieving quality.
- **Methods:** Provide details on how to build software. Typically include requirements and design specification, analysis, coding, testing and maintenance.
- **Tools:** Provide fully automated or partial support for the process and methods.

Three generic phases

- **Definition phase**
- **Development phase**
- **Support phase**

Three generic phases

- **Definition phase:**

- Key requirements of the system and software are defined.
- Typically three major tasks are performed:
 - System engineering
 - Project planning
 - Requirements specification and analysis

- **Development phase**

- **Support phase**

Three generic phases

- **Definition phase**
- **Development phase**
 - Answers all questions related to design, architecture, coding and testing.
 - Typically, three tasks are performed:
 - Software design
 - Coding
 - Testing
- **Support phase**

Three generic phases

- **Definition phase**
- **Development phase**
- **Support phase**
 - Handles change that occurs due to errors in software, changes in customer requirements etc.
 - Tasks done during design and development phases are re-done here, based on the need.

Umbrella activities in SE

- Various phases of SE are supported by several umbrella activities:
 - Project planning, tracking and control
 - Quality assurance
 - Risk assessment and management
 - Reviews
 - Metrics and their measurement
 - Configuration management
 - Documentation
 - ...